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INTER-OFFICE CORRESPONDENCE
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To: . Mr. T. A. Newman

From: . C. L. Irving

Subject: . Evaluation of Borgwaldt Firmness Tester

C81- 05224

Date: July 31, 1981

The evaluation of the Borgwaldt Firmness Tester will determine:

- 1) If the Borgwaldt unit is as sensitive to cigarette firmness as the standard PM compacimer,
- 2) If the variability of the Borgwaldt instrument is comparable to that of the PM compacimeter,
- 3) If the Borgwaldt unit is unduly sensitive to other cigarette characteristics such as circumference or seam orientation, and
- 4) If the unit is easier to operate, set-up and maintain than the standard PM compacimeter.

The sensitivity and variability of the Borgwaldt unit will be determined by testing cigarettes of a single blend at five different weight levels on both the Borgwaldt instrument and the PM compacimeter. This portion of the experiment will be performed three times to ensure the accuracy of the results. Cigarettes will also be tested at two circumference levels and five weight levels to determine the sensitivity of the Borgwaldt unit to circumference variation. In addition, a set of cigarettes will be tested with three different seam orientations to determine the sensitivity of Borgwaldt firmness results to seam position. The data from these experiments will be analyzed to determine the sensitivity and variability of the Borgwaldt instrument so it can be compared with the PM compacimeter.

The experience gained while performing the tests outlined above should provide a good indication of the ease with which the Borgwaldt instrument can be set-up and operated. The determination of the maintainability of the unit is strictly an engineering judgement.

The evaluation of the Borgwaldt firmness tester, as outlined above, should require approximately two months to complete after receipt of the instrument. If you have any questions about this test procedure or any suggestions for additions to the program, please, let me know.

jbn

cc: Dr. W. F. Gannon
Mr. J. S. Osmalov

Christopher L. Irving

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